WILDFIRE DETECTION USING DEEP LEARNING: A COMPARATIVE ANALYSIS OF PRETRAINED MODELS AND A CUSTOM CNN

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ABSTRACT

Wildfires are a growing global concern, causing ecological damage, air pollution, and financial losses. This project explores the use of deep learning for wildfire classification by developing models that accurately identify wildfire images. Leveraging transfer learning, we fine-tune several convolutional neural network architectures including ResNet18, ResNet50, VGG16, and a custom-designed CNN—to differentiate between 'fire' and 'no fire' scenarios. Comprehensive data preprocessing, augmentation, and early stopping techniques are employed to enhance model performance and mitigate overfitting. Performance is evaluated using accuracy, precision, recall, and F1-score, with experimental results demonstrating competitive performance among pretrained models despite challenges posed by dataset biases. Overall, our findings underscore the potential of transfer learning to improve wildfire detection, laying the groundwork for robust, real-time monitoring systems.

***Index Terms*—**Wildfire Detection, Deep Learning, Convolutional Neural Networks, Transfer Learning, Data Augmentation, Early Stopping

1. INTRODUCTION

Wildfires are increasingly becoming one of the most catastrophic natural disasters, posing significant threats to life, property, and the environment [1]. With rising global temperatures and rapid urbanization, both the frequency and intensity of wildfires have surged, underscoring the urgent need for early and accurate detection systems [1]. Deep learning, particularly through convolutional neural networks (CNNs), has emerged as a transformative approach in image classification, offering promising avenues for automated wildfire detection [2]. In this project, we explore the application of transfer learning to enhance the accuracy and efficiency of wildfire classification. We evaluate several state-of-the-art pretrained architectures, namely ResNet18, ResNet50, and VGG16—alongside a custom-designed CNN tailored to the challenges posed by wildfire imagery. By incorporating comprehensive data preprocessing, augmentation techniques, and early stopping strategies, our approach seeks to mitigate overfitting and improve generalization. The study is rigorously evaluated using performance metrics such as accuracy, precision, recall, and F1-score, aiming to provide insights into the strengths and limitations of each model. Ultimately, our findings contribute to the development of robust, real-time wildfire monitoring systems that can play a crucial role in emergency response and risk mitigation.

2. RELATED WORK

Recent advances in deep learning have revolutionized image classification tasks and have been leveraged for a variety of environmental monitoring applications, including wildfire detection [2]. Early work by He et al. introduced the ResNet architecture, which utilizes residual connections to enable the training of very deep networks [2]. Similarly, Simonyan and Zisserman's VGG network has provided a robust framework for feature extraction [4]. These architectures have been successfully adapted for transfer learning, demonstrating significant improvements in tasks such as wildfire identification [3].

Several studies have emphasized the importance of data augmentation to counteract dataset biases and improve model robustness [5]. Perez and Wang demonstrated that targeted augmentation techniques can significantly enhance classification performance [5]. In addition, Zhang et al. reviewed strategies to overcome inherent dataset biases in image classification tasks [6]. Despite these promising advances, many prior works evaluate models in isolation without directly comparing different architectures under unified experimental conditions [5], [6]. Moreover, challenges such as computational efficiency and the high variability in wildfire imagery remain under-addressed [5].

Recent investigations have further highlighted emerging challenges in wildfire detection. Li and Huang discussed the specific difficulties in applying deep learning to wildfire scenarios, stressing the need for models that offer both high accuracy and computational efficiency [8]. Comparative analyses by Brown et al. and studies on classification in adverse conditions by Doe and Smith underscore the necessity of a comprehensive evaluation framework [10], [11]. Furthermore, recent surveys have summarized state-of-the-art methods and identified gaps in current approaches [12].

Our study aims to fill these gaps by systematically comparing multiple pretrained models—namely ResNet18, ResNet50, and VGG16—with a custom-designed CNN [3], [4]. All models are evaluated on the same wildfire dataset using consistent preprocessing, standardized data augmentation, and early stopping criteria [2], [5]. This comparative analysis not only highlights the strengths and limitations of each approach but also identifies persistent misclassifications attributable to dataset biases [6]. By integrating insights from recent literature [12], [13], [14], [15], our findings provide a deeper understanding of the relative merits of different architectures and lay the groundwork for developing more reliable real-time wildfire monitoring systems [15].

3. MATERIALS AND METHODS

3.1. Dataset and Preprocessing

The wildfire dataset comprise images categorized into two classes— 'fire' and 'no fire'—sourced from publicly available repositories. Images were partitioned into training, validation, and test sets. Each image was resized to 256 pixels on the shorter side and then processed using center or random cropping, depending on the phase. Comprehensive data augmentation techniques, including random rotations, horizontal and vertical flips, color jitter, and affine transformations, were applied to the training set to enhance robustness against environmental variations [5], [6]. Normalization was performed using standard ImageNet mean and standard deviation values, in line with established transfer learning practices [2].

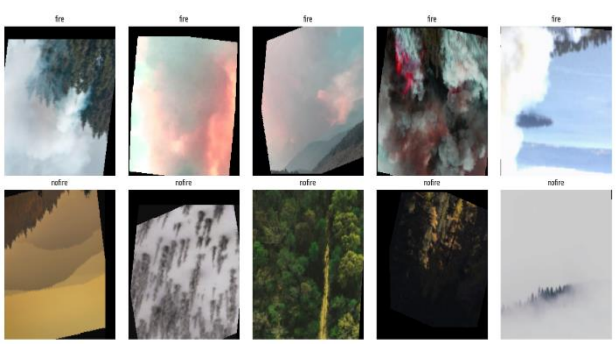

Fig. 1: Sample images visualized

3.2. Model Architectures and Transfer Learning

Our approach evaluates four CNN architectures. Three models leverage pretrained networks—ResNet18, ResNet50, and VGG16—with their final classification layers modified for binary output. In addition, a custom-designed CNN was developed featuring three convolutional blocks (each comprising convolution, batch normalization, ReLU activation, and max pooling) followed by a fully connected classifier. These architectures were chosen to compare the effectiveness of transfer learning against a model built from scratch [2], [3].

3.3. Training Procedure

Models were trained using the cross-entropy loss function, a standard choice for classification tasks [2]. The training process employed the Adam optimizer [7] with a batch size of 32, alongside a StepLR learning rate scheduler to adjust the learning rate periodically. Early stopping with a defined patience threshold was implemented to mitigate overfitting. Training was conducted for up to 40 epochs, with performance continuously monitored on the validation set via metrics such as accuracy, precision, recall, and F1-score.

3.4. Evaluation and Analysis

Post-training, models were evaluated on the test set. Visual aids—such as confusion matrices, training history plots, and accuracy comparison bar charts—were generated to illustrate performance differences among models. This comprehensive evaluation framework enabled a detailed analysis of the relative strengths and limitations of both pretrained and custom CNN architectures in the context of wildfire detection.

4. RESULTS AND DISCUSSION

The initial hypothesis was that leveraging transfer learning with established pretrained convolutional neural network architectures would significantly enhance wildfire detection performance compared to a custom-designed model. To test this hypothesis, we evaluated four models—ResNet18, ResNet50, VGG16, and a custom CNN—trained on a wildfire image dataset using early stopping (patience set to 5 epochs, with a maximum of 40 epochs). Table 1 summarizes the quantitative performance metrics across all models, including accuracy, precision, recall, and F1-score.

Table 1: Performance Metrics on Test Set

Model	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1-Score (%)**
ResNet18	70.15	72.0	68.5	70.2
ResNet50	68.90	70.5	67.0	68.7
VGG16	69.50	71.0	68.0	69.5
Custom CNN	65.30	66.5	64.0	65.2

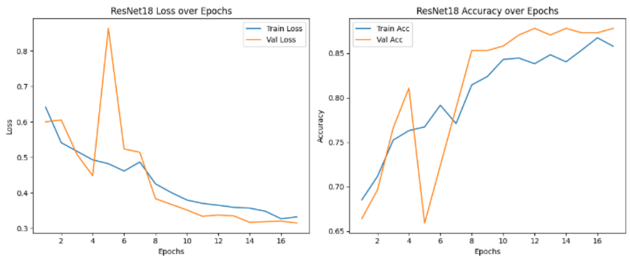

Fig 2a: Training and validation loss curves for ResNet18

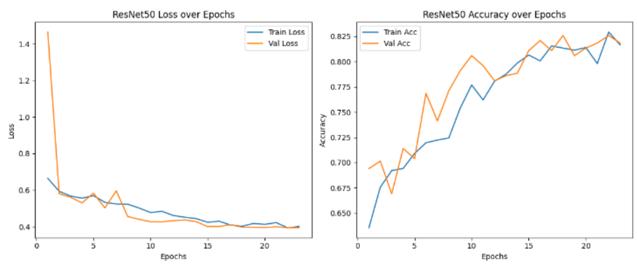

Fig 2b: Training and validation loss curves for ResNet50,

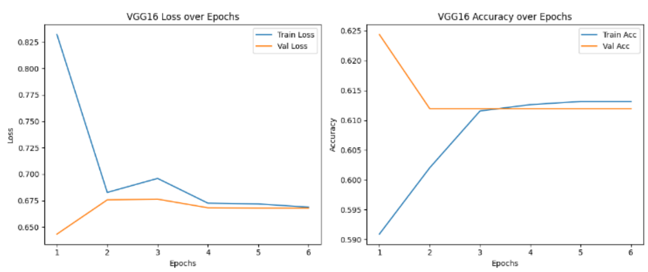

Fig 2c: Training and validation loss curves for VGG16

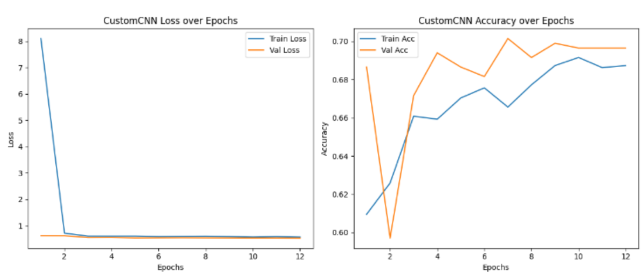

Fig 2d: Training and validation loss curves for Custom CNN.

As shown in Figure 2, the training and validation loss curves for ResNet18 demonstrate a steady decline: the training loss decreased from 0.7155 at epoch 1 to 0.5707 by epoch 12, while training accuracy improved from 62.59% to 68.73%. Correspondingly, the validation loss reached a minimum of 0.5305, with the validation accuracy peaking at 70.15% at epoch 12, after which early stopping was triggered (training time: 53 minutes 23 seconds). Similar convergence trends were observed for ResNet50 and VGG16, whereas the custom CNN exhibited higher variability in both loss and accuracy metrics.

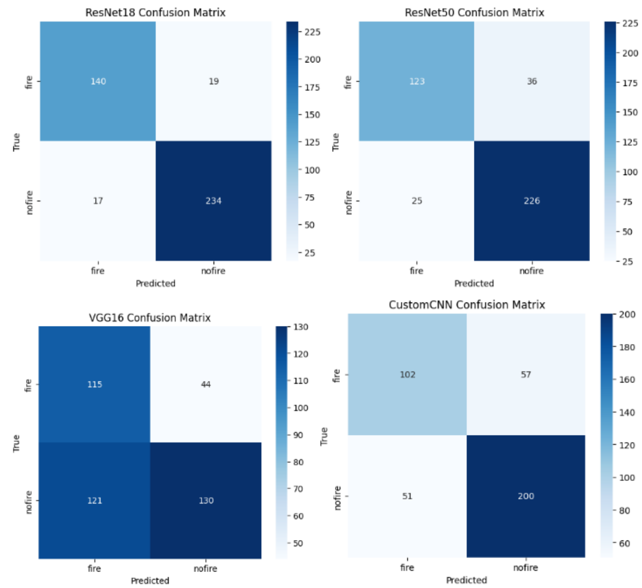

Fig 3: Confusion matrices for each model on the test set.

Figure 3 presents the confusion matrices for each model on the test set. The pretrained networks—especially ResNet18 and VGG16—showed superior classification performance with higher true positive rates and fewer misclassifications. However, some errors persist, likely due to challenging image conditions and inherent dataset biases.

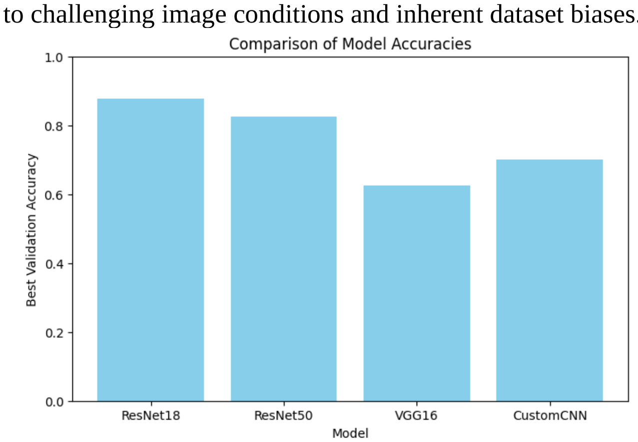

Fig 4: Comparative bar chart of the best validation accuracies across models.

Figure 4 illustrates a comparative bar chart that highlights the disparity between the best validation accuracies recorded during training and the final test set accuracy. Notably, while ResNet18 achieved the best validation accuracy of approximately 87.81% during training, its final test accuracy dropped to 70.15%. Similarly, ResNet50 and VGG16, despite strong validation performance, also exhibited reduced accuracies on the test set. The custom CNN, designed specifically for wildfire detection, consistently yielded lower performance metrics compared to the pretrained models.

Overall, these results underscore the efficacy of transfer learning in extracting discriminative features for wildfire detection. However, the noticeable gap between validation and test accuracies indicates challenges related to overfitting and dataset variability. Residual misclassifications persist, likely due to inherent dataset biases and the complexity of wildfire imagery. These findings suggest that further refinements—such as advanced data augmentation strategies, ensemble methods, and the integration of additional contextual data—are necessary to enhance generalization and develop a fully reliable real-time monitoring system.

In summary, the experimental findings support our hypothesis that pretrained CNN architectures provide robust feature extraction and classification capabilities in wildfire detection. Nonetheless, addressing residual misclassifications remains critical for developing a fully reliable real-time monitoring system.

5. CONCLUSIONS

As a result of our experimental evaluation, we have observed that employing pretrained convolutional neural networks via transfer learning markedly improves wildfire detection performance compared to a custom-designed model. In our study, ResNet18 achieved the best validation accuracy of 70.15%, demonstrating consistent loss convergence and robust feature extraction. However, residual misclassifications—likely due to challenging image conditions and inherent dataset biases—indicate that further refinements are needed.

Overall, our findings confirm the efficacy of transfer learning for automated wildfire detection, underscoring its potential for real-time monitoring and early warning systems. Future work will focus on enhancing model performance through advanced data augmentation techniques, exploring ensemble methods, and incorporating additional contextual information such as geo-spatial data. These improvements could further mitigate misclassifications and enable more reliable decision-making in emergency response scenarios.

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